

ZetaGo

by Team Sharp Features

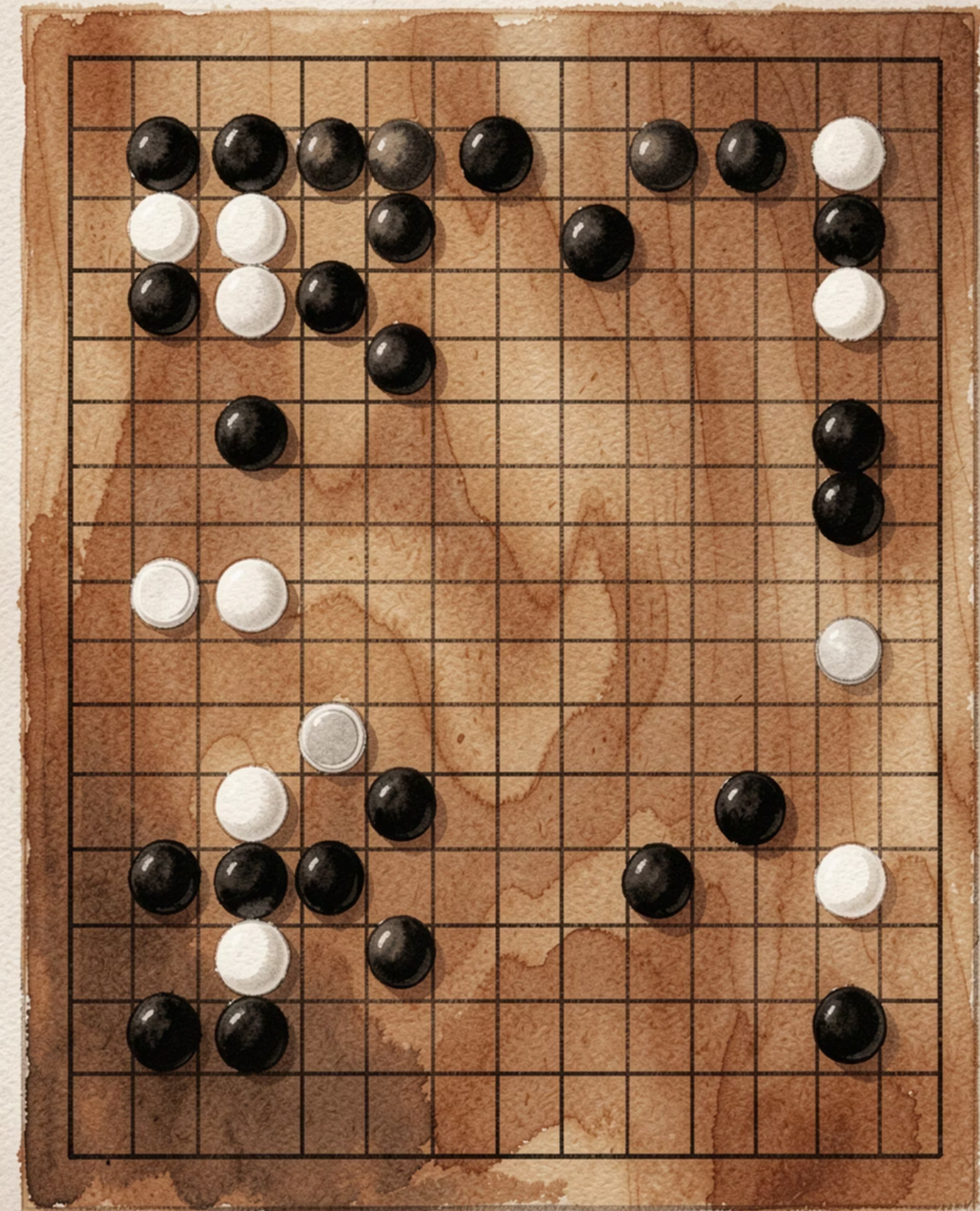
Feature Representation & Warm-Start Strategy in AlphaZero-Style Self-Play on 7×7 Go

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MOTIVATION: WHY THIS PROBLEM?

AlphaGo Zero proved a self-play policy network can surpass human-level Go, but it required **hundreds of TPUs running for months**.

Under **bounded compute and limitations**, which design decisions actually matter?

- How many input feature planes do you actually need?
- Does supervised pre-training help or can you start cold?
- How fast do agents converge under matched compute?

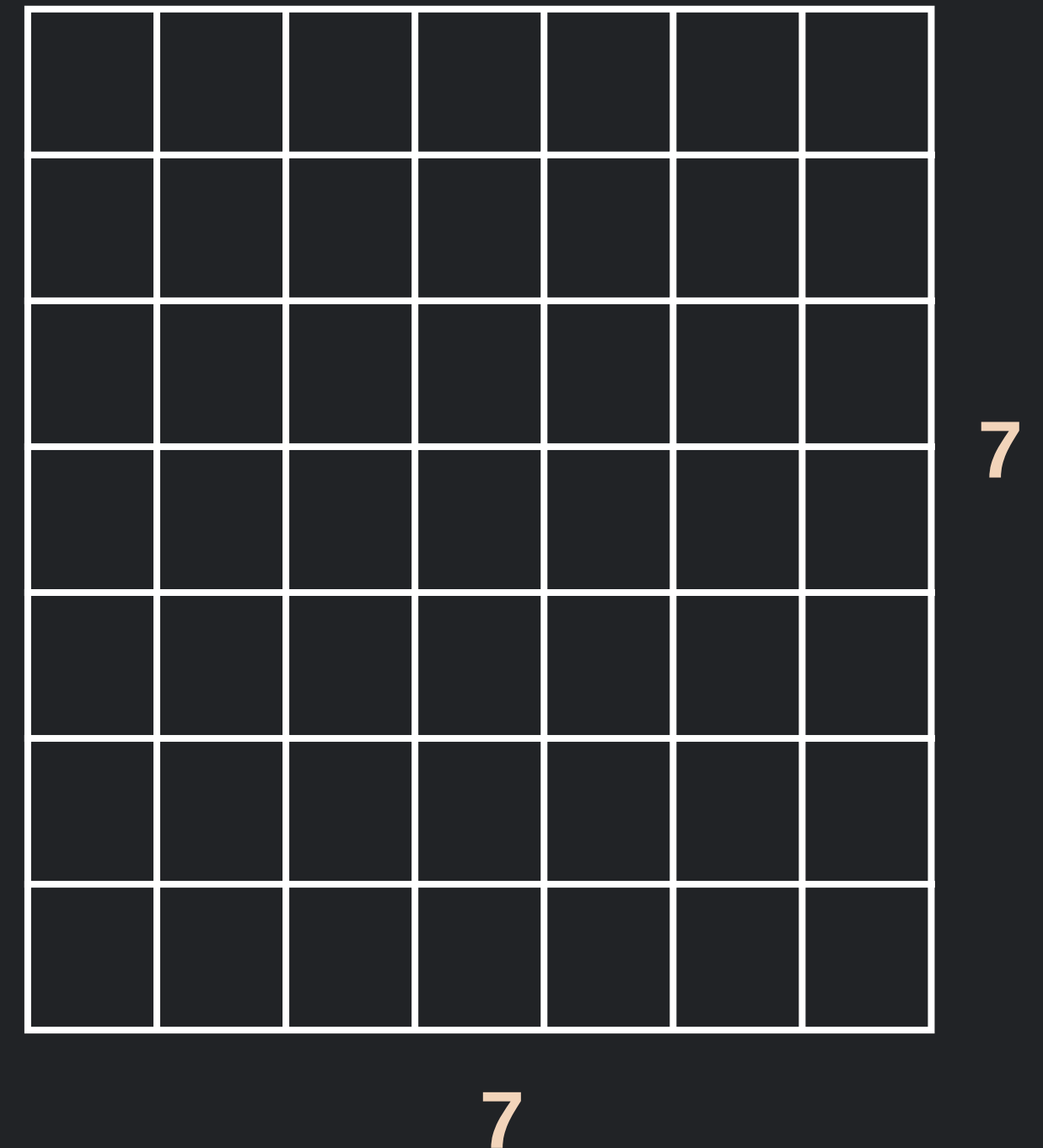


THE GAP

No published study has run a factorial of **input feature-plane count × warm-start strategy** on 7×7 Go from our studies.

Our scope: A controlled **2×3 factorial experiment** isolating feature richness and initialization strategy — the first of its kind at this board size.

7×7 Go is feasible for smaller scale experiments while still retaining the rich tactical structure of Go.



RESEARCH QUESTION

Does the richness of input feature representation (N = 2 / 4 / 7 planes) interact with supervised pre-training strategy to affect the sample efficiency and final playing strength of a self-play MCTS agent on 7×7 Go?

2 × 3 Factorial Design

	N = 2 Minimal	N = 4 Standard	N = 7 Rich
Cold-Start RL	C1	C2	C3
Supervised Pre-train	C4	C5	C6

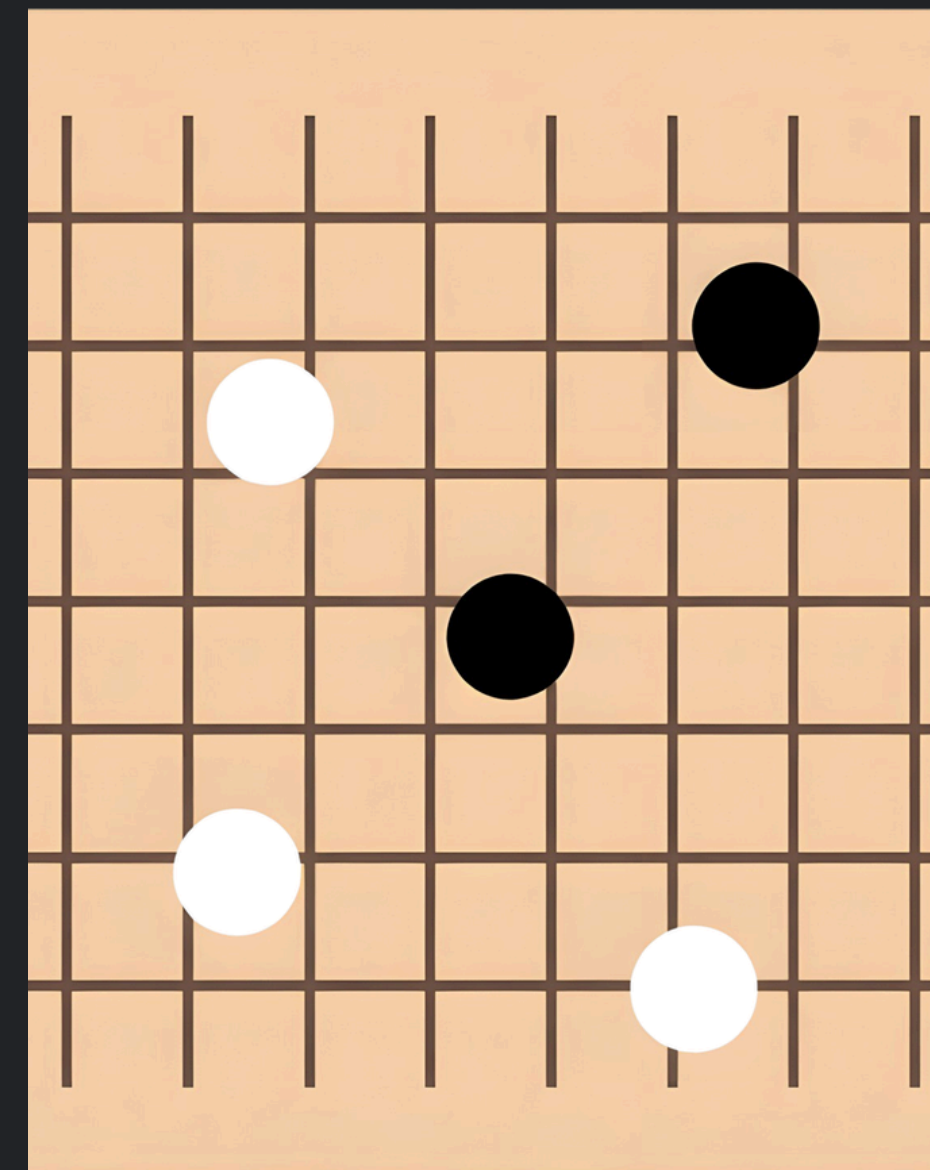
RELATED WORK: WHAT HAS BEEN DONE BEFORE?

Ref	Paper	What They Studied	Our Gap
R1	AlphaGo · Silver et al. · 2016	Defined feature planes for Go	19×19 only, no feature-count ablation
R2	AlphaGo Zero · Silver et al. · 2017	Pure self-play, minimal features	No warm-start vs. cold-start comparison
R8	KataGo · Wu · 2019	NoGoFeat ablation: removes feature block	Single block removal, not graded plane count
R5	Soemers et al. · 2019	Feature biasing in MCTS — general games	Linear features only, not deep CNNs
R6	Wang et al. · 2020	Hyper-parameter sweep on small-board games	Varies hyperparameters, not input plane count

DATASET: KATAGO SELF-PLAY

Source: KataGo self-play (CPU Eigen backend)

- **Size:** 50,000–100,000 game records
- **Format:** (board tensor, move, game outcome) triples
- **Storage:** NumPy / HDF5
- KataGo runs in background from Day 1 (2–4 days generation time).



N=2 · MINIMAL

- Current stones (black)
- Current stones (white)



N=4 · STANDARD

- + Liberty count
- + Turn indicator



N=7 · RICH

- + Ladder status
- + Ko point
- + Move history (2 steps)

SAMPLE DATASET

```

1 game_id,move_no,to_move,board,move,move_index,value,margin,winner
2 3765209461,0,B,...../...../...../...../...../...../...../.....,"3,3",24,-1,-0.5,W
3 3765209461,1,W,...../...../...../...X.../...../...../...../.....,"4,3",31,1,0.5,W
4 3765209461,2,B,...../...../...../...X.../...0.../...../...../.....,"3,4",25,-1,-0.5,W
5 3765209461,3,W,...../...../...../...XX.../...0.../...../...../.....,"4,2",30,1,0.5,W
6 3765209461,4,B,...../...../...../...XX.../...00.../...../...../.....,"3,2",23,-1,-0.5,W
7 3765209461,5,W,...../...../...../...XXX.../...00.../...../...../.....,"4,4",32,1,0.5,W
8 3765209461,6,B,...../...../...../...XXX.../...000.../...../...../.....,"4,1",29,-1,-0.5,W
9 3765209461,7,W,...../...../...../...XXX.../...X000.../...../...../.....,"5,1",36,1,0.5,W
10 3765209461,8,B,...../...../...../...XXX.../...X000.../...0.../...../.....,"3,1",22,-1,-0.5,W
11 3765209461,9,W,...../...../...../...XXXX.../...X000.../...0.../...../.....,"4,5",33,1,0.5,W
12 3765209461,10,B,...../...../...../...XXXX.../...X0000.../...0.../...../.....,"3,5",26,-1,-0.5,W
13 3765209461,11,W,...../...../...../...XXXXX.../...X0000.../...0.../...../.....,"3,6",27,1,0.5,W
14 3765209461,12,B,...../...../...../...XXXXX0/.X0000.../...0.../...../.....,"2,6",20,-1,-0.5,W
15 3765209461,13,W,...../...../...../...X/.XXXXX0/.X0000.../...0.../...../.....,"4,6",34,1,0.5,W
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17 3765209461,15,W,...../...../...../...XX/.XXXXX0/.X00000/.0.../...../.....,"5,0",35,1,0.5,W
18 3765209461,16,B,...../...../...../...XX/.XXXXX0/.X00000/00.../...../.....,"4,0",28,-1,-0.5,W
19 3765209461,17,W,...../...../...../...XX/.XXXXX0/XX00000/00.../...../.....,pass,49,1,0.5,W
20 3765209461,18,B,...../...../...../...XX/.XXXXX0/XX00000/00.../...../.....,pass,49,-1,-0.5,W
21 2540419555,0,B,...../...../...../...../...../...../...../.....,"3,3",24,-1,-0.5,W
22 2540419555,1,W,...../...../...../...X.../...../...../...../.....,"4,3",31,1,0.5,W
23 2540419555,2,B,...../...../...../...X.../...0.../...../...../.....,"3,2",23,-1,-0.5,W
24 2540419555,3,W,...../...../...../...XX.../...0.../...../...../.....,"4,4",32,1,0.5,W

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EXPECTED CONTRIBUTIONS

Concrete deliverables across empirical, classical, and unsupervised ML tracks that we expect ZetaGo to produce:

1

Primary Empirical Finding

First reported interaction between input-feature richness $\times$ supervised warm-start under bounded compute on 7 $\times$ 7 Go — 2 $\times$ 3 factorial, 36 training runs, 95% bootstrap CIs.

3

Unsupervised Position Analysis

Convolutional autoencoder on 50k board positions + PCA + k-Means. Do unsupervised representations encode tactically meaningful structure?

2

Classical ML Baselines

Logistic Regression, Random Forest, k-NN trained as MCTS policy priors — compute-matched comparison against shallow and deep CNNs.

4

Comparative Representation Analysis

CNN activations (post-RL) vs. autoencoder latents — do supervised/RL-trained representations organize positions differently?

TOOLS & FRAMEWORKS: TECHNICAL STACK



PyTorch

CNN policy-value network with configurable input planes



scikit-learn

LR · RF · k-NN baselines · PCA · k-Means clustering



BayesElo

Elo rating computation from round-robin tournaments



KataGo

7×7 game record generation via CPU Eigen backend



Google Colab T4

Primary training platform for all experiments



Matplotlib / Seaborn

Convergence curves · cluster visualizations · paper figures

CHALLENGES & RISK MITIGATION

Hardware Uncertainty

University GPU access not confirmed. CPU-only may limit scale.

→ **Mitigation:** Design first for CPU / 5×5. Scale to 7×7 only with GPU. Every step has a fallback.

Seed Noise > Effect Size

At small compute budgets, variance between runs may swamp the true signal.

→ **Mitigation:** 3+ seeds per condition. Report 95% bootstrap CIs. Pre-register on OSF.

RL Convergence Failure

Self-play may not converge within the compute budget.

→ **Mitigation:** SL-only study is a complete standalone paper. RL is an extension, not a requirement.

Game Engine Bug

A subtle Tromp-Taylor scoring or Ko rule error could silently corrupt all training data.

→ **Mitigation:** Comprehensive unit tests before any ML work begins.

KataGo Dataset Latency

CPU generation may be slow, creating a pipeline bottleneck.

→ **Mitigation:** Start KataGo on Day 1 in background — fully independent of all other tasks.

References

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[R5]

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THANK YOU